Unit: Unit 2: Exponents Radicals
Topic: Rationalizing Denominator

Name: ________________
Date: ___________

Quiz (Carolina Reaper)

Q1. A shipping crate's packing density is defined by the ratio $\frac{120}{\sqrt{5}+\sqrt{2}}$. Which expression represents the rationalized density in kg/m^3?
(A) $40\sqrt{5}-40\sqrt{2}$
(A) $60\sqrt{5}-60\sqrt{2}$
(A) $40\sqrt{7}$
(A) $20(\sqrt{5}-\sqrt{2})$
(A) $40\sqrt{3}$

Q2. The flow rate of a logistics pipeline is modeled as $F = \frac{15}{\sqrt{10}-\sqrt{7}}$. Simplify this for the flow control software.
(A) $5\sqrt{10}-5\sqrt{7}$
(A) $15\sqrt{17}$
(A) $5\sqrt{10}+5\sqrt{7}$
(A) $3\sqrt{70}$
(A) $5(\sqrt{3})$

Q3. A delivery drone's average speed over a windy canyon is $\frac{20}{4-\sqrt{6}}$ km/h. What is its speed in simplest radical form?
(A) 8 + $2\sqrt{6}$
(A) 4 + $\sqrt{6}$
(A) 2 + $8\sqrt{6}$
(A) 8 - $2\sqrt{6}$
(A) 10 + $2\sqrt{6}$

Q4. The friction coefficient of a conveyor belt is calculated as $\mu = \frac{7\sqrt{3}}{2\sqrt{3}+\sqrt{5}}$. Rationalize μ.
(A) 6 - $\sqrt{15}$
(A) $7(\sqrt{3}-\sqrt{5})$
(A) 42 - $7\sqrt{15}$
(A) 6 + $\sqrt{15}$
(A) 42 - $7\sqrt{15}_{\overline{7}}$

Q5. An automated warehouse lift experiences a tension force of $\frac{42}{\sqrt[3]{4}}$ Newtons. Rationalize the denominator.
(A) $21\sqrt[3]{4}$
(A) $10.5\sqrt[3]{16}$
(A) $21\sqrt[3]{2}$
(A) $42\sqrt[3]{2}$
(A) $14\sqrt[3]{4}$

Q6. The time delay at a harbor checkpoint is $T = \frac{1}{\sqrt{x}+\sqrt{x-1}}$ hours. Which expression is equivalent?
(A) $\sqrt{x}+\sqrt{x-1}$
(A) $\sqrt{x}-\sqrt{x-1}$
(A) $2\sqrt{x}-1$
(A) $\sqrt{x}-\sqrt{x-1}_{\overline{2x-1}}$
(A) 1

Q7. A cargo ship's fuel efficiency ratio is given by $\frac{\sqrt{x}+\sqrt{y}}{\sqrt{x}-\sqrt{y}}$. Find the rationalized form.
(A) x + $\text{y}_{\overline{x-y}}$
(A) x + $2\sqrt{xy}+y_{\overline{x-y}}$
(A) 1
(B) x - $2\sqrt{xy}+y_{\overline{x-y}}$
(B) $\sqrt{x^2+y^2}_{\overline{x-y}}$

Q8. Optimal routing distance in a complex grid is $\frac{10}{3\sqrt{2}-4}$. What is the simplified distance?
(A) $15\sqrt{2}+20$
(A) $15\sqrt{2}-20$
(A) $30\sqrt{2}+40$
(A) $5\sqrt{2}+4$
(A) $10\sqrt{2}+10$

Open Ended Questions (FRQ)

Q9. A cargo plane's lift-to-drag coefficient involves the term $\frac{34}{3\sqrt{5}-\sqrt{11}}$. Rationalize the denominator and simplify fully.

Q10. A traffic throughput algorithm requires simplifying the expression $P = \frac{x-16}{\sqrt{x}-4} + \frac{x-y}{\sqrt{x}+\sqrt{y}}$ for $x, y > 0$. Provide the fully simplified, rationalized expression.

Chef's Tips

- Verify that students multiply by the exact conjugate, focusing on sign changes.
- Watch for errors in squaring binomials in the numerator for FRQ 10.
- Ensure students understand that $\sqrt[3]{a}$ requires $\sqrt[3]{a^2}$ to rationalize.